

1. THE 60-SECOND PLAIN-ENGLISH TRANSLATION

WHY DOES THIS MATTER TO SOCIETY?

If you are not an expert in government bureaucracy, public procurement can look like an intimidating alphabet soup of acronyms (GFR, EMD, DFPR, NICGEP, BOQ). Here is the simple, real-world reality of what is happening and why we built CHEIRAP AI (চাইরাপ).

WHAT IS A GOVERNMENT TENDER IN PLAIN ENGLISH?

When the government wants to build a school, pave a highway, or buy medical equipment for a hospital, it cannot just hire a friend's company. By law, it must post a public job notice—called a **Tender**—on an online portal (like an official government marketplace). Any qualified company should be allowed to submit a competitive price proposal (called a **Bid**). The honest company with the best price and quality wins the contract.

💡 THE REAL-WORLD METAPHOR: THE BANK ROBBERY VS. THE VAULT METAL DETECTOR

The Old Way (Traditional Auditing): Imagine a bank that only checks its security cameras *two years after* a robbery occurred. By then, the vault is empty, the corrupt contractors have fled with a 15% upfront cash advance, and taxpayers are left with an unfinished, broken bridge. That is what a "Post-Award Audit" is—an autopsy of money already lost.

The CHEIRAP Way (Pre-Award Interception): CHEIRAP is a smart security scanner installed right at the bank entrance. *Before* any contract is awarded, CHEIRAP analyzes the paperwork in real-time. If it detects that an insider is trying to rig the deal, it alerts the Vigilance Commissioner within 48 hours to freeze the process before the vault door opens.

The "Procurement Dictionary" for Judges

BUREAUCRATIC JARGON	WHAT IT ACTUALLY MEANS	WHY CORRUPT ACTORS ABUSE IT
Bidding Window (Notice Period)	The number of days an opportunity remains open online for vendors to see it and apply.	The Flash Sale Trick: If an officer wants their cousin to win, they secretly publish the notice for just 3 days instead of the mandatory 14–21 days. Honest businesses never even see it.
Corrigendum	An official amendment or correction published to change tender terms after release.	The Midnight Syllabus Change: Sneaking in a drastic technical requirement 12 hours before deadline so competitors are disqualified, leaving only the favored insider.
EMD (Earnest Money Deposit)	A refundable security deposit a company must pay upfront just to submit a proposal.	The High Cover Charge: Demanding an absurdly high deposit (e.g. ₹50 Lakhs for a small job) to choke the cash flow of local small startups and MSMEs.
DFPR (Financial Delegation)	Legal spending limits defining who is allowed to sign off on what budget amount.	Split Tendering: Taking a ₹15 Crore project and chopping it into four ₹3.75 Crore tenders so a lower-level officer can approve it without cabinet oversight.
Single-Bidder Cartel	When only one solitary company submits a bid for a public contract.	The Solo Walkover: Writing tender specifications so bizarrely specific that only one friend qualifies, and the government is forced to pay whatever monopoly price they charge.

TOTAL PUBLIC CAPEX
MONITORED

HIGH-RISK CAPEX INTERCEPTED

PRE-AWARD RESPONSE
WINDOW

LITIGATION AVOIDANCE
RATE

Trick 1: "The Midnight Flash Sale" (Window Compression)

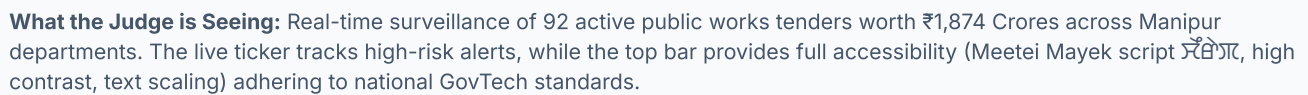
CHEIRAP DETECTION The AI calculates `window_hours = (closing_date - publish_date)`. If it falls below 336 hours (14 days), an immediate statutory red penalty (+35 pts) is triggered.

CHEIRAP DETECTION The engine monitors the portal's corrigenda feed. If $(\text{closing_time} - \text{corrigendum_time}) < 48$ hours without a 7-day extension, it flags a CVC violation (+20 pts).

CHEIRAP DETECTION The model calculates $\text{emd_ratio} = \text{emd_amount} / \text{estimated_contract_value}$. Any ratio above 5.0% triggers an instant alert for anti-competitive exclusion.

CHEIRAP DETECTION bidder_count == 1 immediately flags the case as requiring mandatory market benchmarking before any work order can be generated.

MANIPURTENDERS.GOV.IN INGESTION



WHY NOT JUST USE CHATGPT?

⚠️ WHY LARGE LANGUAGE MODELS (LLMs) FAIL IN GOVERNMENT PROCUREMENT

- 1. LLMs Hallucinate:** A Large Language Model generates probabilistic text. You cannot legally cancel a ₹50 Crore hospital construction project in a High Court of Law by saying "the AI chatbot had a feeling it was rigged."
- 2. Judicial Defensibility:** Indian courts require *deterministic, mathematically reproducible evidence*. If an IAS officer signs a Stay Order, the evidence must stand up under cross-examination.
- 3. Complex Numerical Dimensions:** Cartel patterns are hidden in dates, ratios, capex bands, and bidder cross-correlations—mathematical problems that LLMs routinely miscalculate.

CHEIRAP combines two specialized engines that balance each other perfectly:

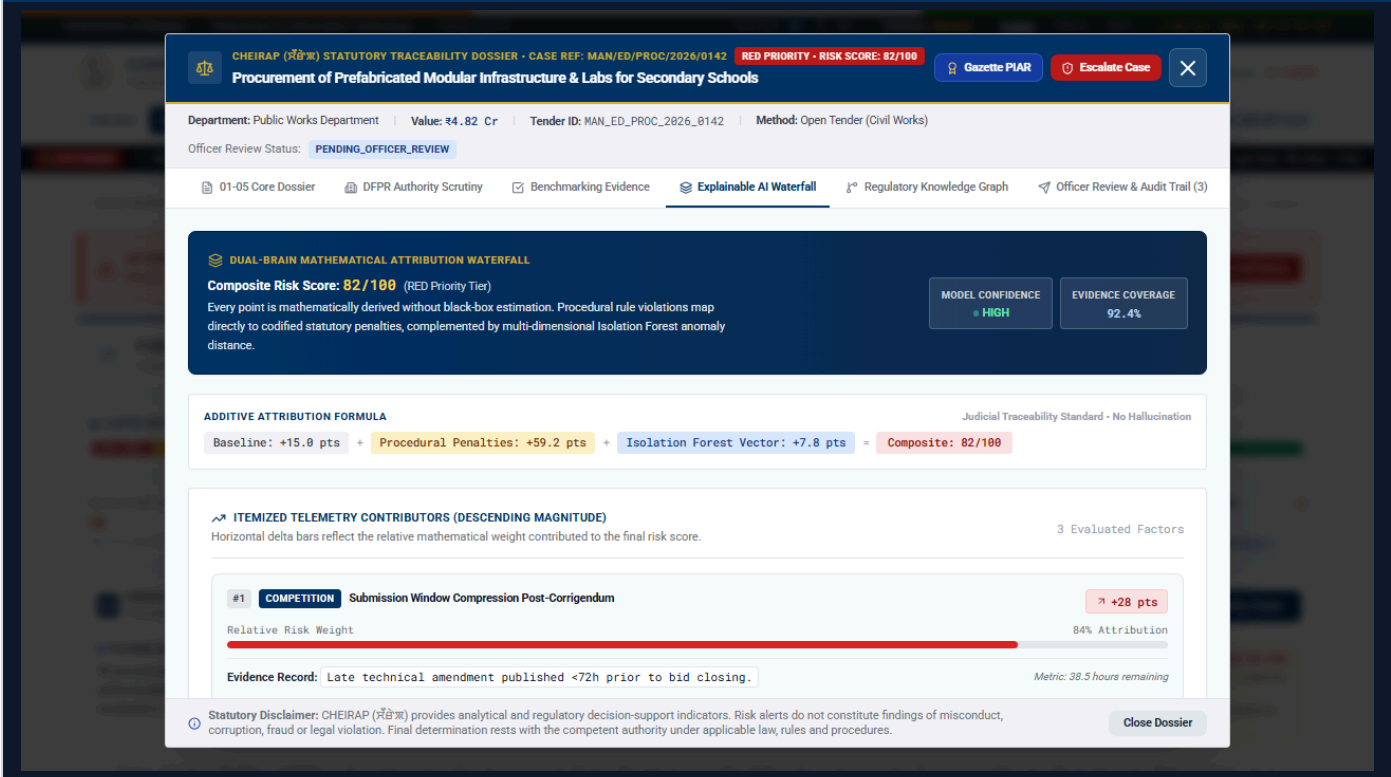
THE FINAL FORMULA

Composite Risk Score = 0.50 × (ML Anomaly Score) + 0.50 × (Statutory Rule Penalty)

- **GREEN (0–39):** Normal tender, fully compliant.
- **AMBER (40–69):** Procedural advisory; recommend issuing an extension corrigendum.
- **RED (70–100):** Critical integrity risk; warrants immediate pre-award administrative stay.

FIGURE 2: EXPLAINABLE AI (XAI) ATTRIBUTION WATERFALL

THE "ITEMIZED RECEIPT" OF FRAUD



What the Judge is Seeing (The Itemized Receipt): Instead of giving a mysterious score, CHEIRAP generates this mathematical waterfall. It proves to the officer exactly where the points came from: *Submission Window Compression* (+28 pts), *Late Corrigendum* (+18 pts), and *High EMD Ratio* (+12 pts). This is 100% explainable and defensible in court.

WHY GENERIC NATIONAL TOOLS FAIL

The CHEIRAP Innovation: Two-Layer Statutory Knowledge Base

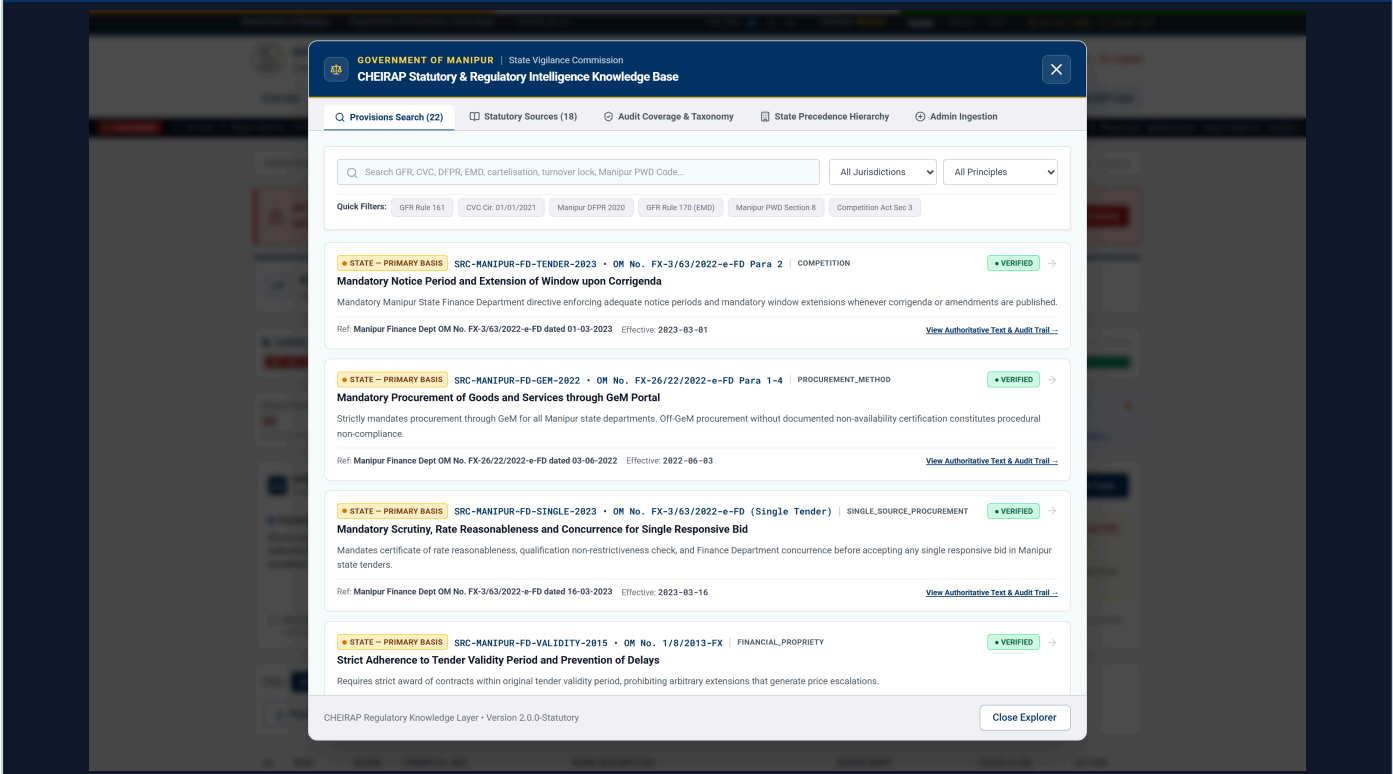
CHEIRAP is the first procurement intelligence system in India designed with a **Two-Layer Legal Hierarchy**:

Layer 1 (State Supremacy): Governed primarily by the **Manipur Finance Department Acts & Rules**. If Manipur rules specify a 14-day window or a specific single-bid benchmarking process, that rule takes absolute precedence over central guidelines.

Layer 2 (Central Baseline): Central GFR 2017 and CVC Vigilance Manual guidelines act as fallback standards for areas where state orders are silent.

OFFICIAL MANIPUR DOCUMENT	WHAT THE GOVERNMENT ORDERED	HOW CHEIRAP ENFORCES IT
OM No. FX-3/63/2022-e-FD (1 March 2023)	Mandatory tender rules for works & services across all Manipur state departments.	Automated checking of minimum bidding days (14–21 days) and tender fee ceilings.
OM No. FX-26/22/2022-e-FD (3 June 2022)	Mandatory procurement of common goods & services via GeM (Government e-Marketplace).	Flags any department attempting off-portal custom tenders for standard goods available on GeM.
OM No. FX-4/2/2023-e-FD (14 July 2023)	Strict protocols for handling single bids: requires market rate comparison & Finance Secretary clearance.	Intercepts tenders where bidder_count == 1 to prevent illegal financial contract awards.
OM No. 1/8/2013-FX (28 February 2015)	Strict limitations on tender validity extensions without administrative approval.	Monitors aging tenders to stop endless, unauthorized validity extensions.
Manipur DFPR 2020 (Delegation of Financial Powers)	Codified financial sanction limits: Executive Engineer (₹50L), Superintending Engineer (₹5 Cr), Chief Engineer (₹25 Cr).	Flags any tender whose contract value exceeds the statutory sanction limit of the issuing officer.

FIGURE 3: THE REGULATORY INTELLIGENCE & KNOWLEDGE BASE EXPLORER TWO-LAYER STATUTORY TRACEABILITY



What the Judge is Seeing: Officers can search and cross-reference official Manipur Finance Department notifications alongside Central GFR clauses in real-time. Every flagged anomaly is linked to an exact clause and legal authority.

5. THE ACTION WEAPON: 14-SECTION GAZETTE REPORT

FROM AI INSIGHT TO ENFORCEABLE STAY ORDER

Most hackathon AI projects end with a dashboard chart. But a dashboard chart cannot stop a rogue contractor. To bridge the gap between software and real-world executive power, CHEIRAP generates a **14-Section Gazette-Formatted Pre-Award Integrity Assessment Report (PIAR)**.

THE "ARREST WARRANT" FOR RIGGED TENDERS

Think of the PIAR as a legal search warrant or stay order. It is formatted in the historic styling of **The Manipur Gazette (Extraordinary)**. It features the official State Emblem, Meetei Mayek script, a formal Gazette reference number, an itemized table of legal violations, and a **cryptographic SHA-256 digital fingerprint**. A Vigilance Commissioner can review it in 2 minutes, hit "Print", and issue an official stay order under **Section 30 of the CVC Vigilance Manual** to freeze the tender before money is paid.

Structure of the 14 Forensic Gazette Sections

GAZETTE SECTION	WHAT IT DOCUMENTS	JUDICIAL & LEGAL SIGNIFICANCE
Sec 01–03: Identity & Capex	Tender ID, department, estimated contract value, and issuing officer hierarchy.	Establishes exact jurisdictional authority and whether value was split to bypass sanctions.
Sec 04–06: Procedural Chronology	Publish date, closing date, bidding window hours, and corrigenda timestamps.	Mathematically proves whether the 14-day window was illegally compressed.
Sec 07–09: Competition & Fees	EMD amount, tender fee, MSME exemption status, and bidder count.	Documents evidence of market foreclosure and artificial entry barriers.
Sec 10–12: AI Attribution & Rules	Isolation Forest anomaly score and itemized statutory rule penalties.	Provides the transparent mathematical reasoning required for judicial scrutiny.
Sec 13–14: Legal Citations & Order	Explicit citations (Manipur OMs + GFR) and formal recommended executive action.	Provides the ready-to-sign legal text for the Chief Vigilance Officer.

What the Judge is Seeing: The standalone Gazette Print View with official Kangla Sha emblem, Meetei Mayek header, bilingual statutory citations, and cryptographic verification hash. In one click, officers can print an official legal stay order.

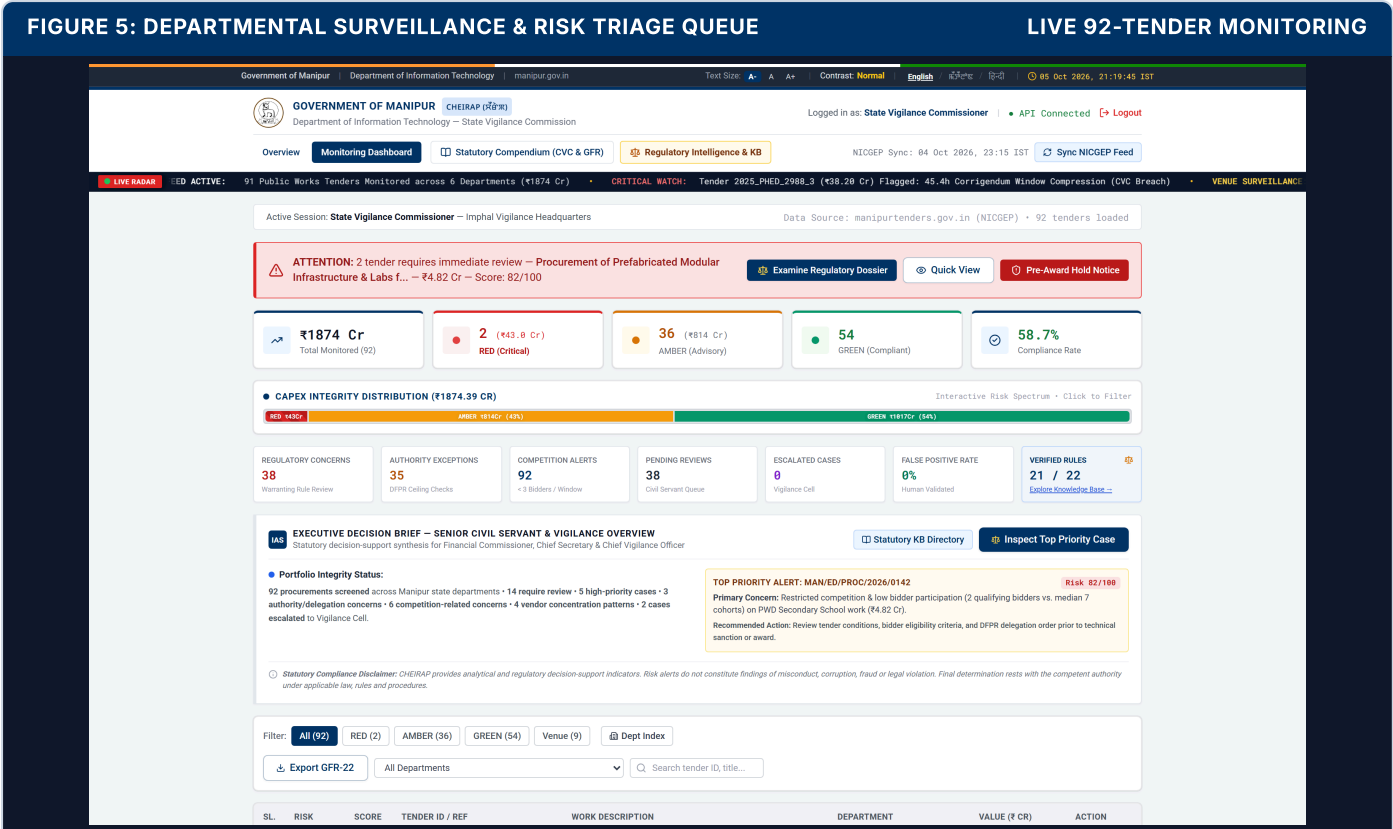
6. LIVE WALKTHROUGH & SCREENSHOT TOUR

HOW AN OFFICER ACTUALLY USES CHEIRAP

Here is the step-by-step user journey of an anti-corruption vigilance officer using CHEIRAP to catch a live rigged tender:

Step 1: Departmental Surveillance & Triage Queue

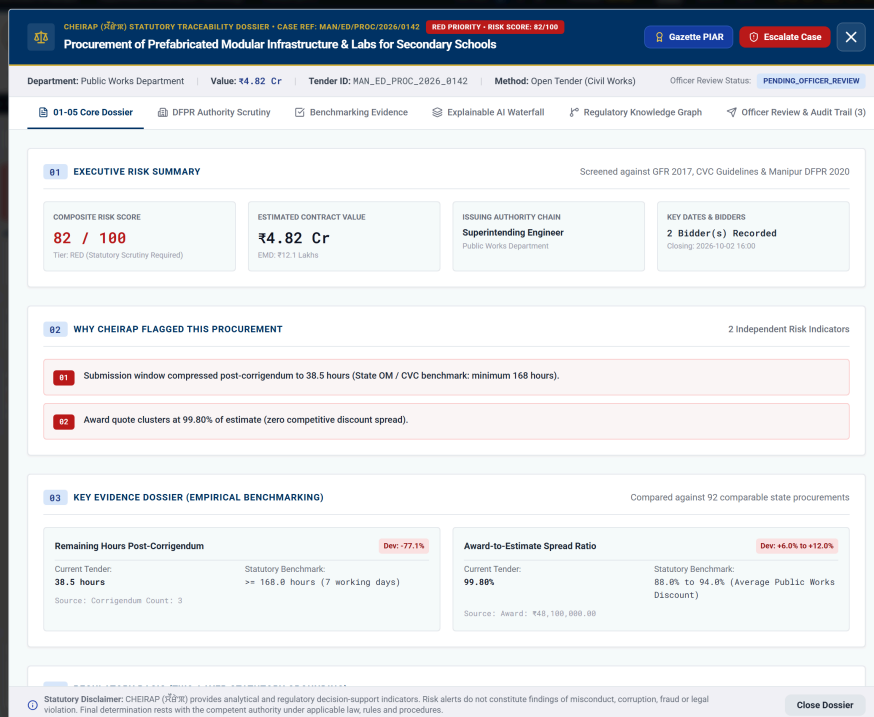
The officer opens the CHEIRAP dashboard. Instead of digging through hundreds of PDF pages on the slow state portal, CHEIRAP displays an interactive queue of 92 tenders triaged by risk: **RED (Immediate Interception)**, **AMBER (Advisory)**, and **GREEN (Clean)**.



What the Judge is Seeing: Instant risk breakdown: 2 Critical Red tenders (₹43.0 Cr capex flagged), 36 Amber advisories, and 54 Clean tenders. Officers can filter by department (Education, PWD, Health) with a single click.

Step 2: Deep Forensic Case Dossier

The officer clicks on the high-risk Education Department tender (MAN_ED_PROC_2026_0142, ₹4.82 Crores for Prefabricated Modular Labs). A comprehensive modal opens with 7 forensic tabs: Executive Summary, DFPR Limit Scrutiny, Evidence Benchmarking, XAI Waterfall, Knowledge Graph, Officer Review Audit Trail, and the 14-Section Gazette PIAR.



What the Judge is Seeing: The complete evidence dossier. Notice the clear badges, contractual timeline, issuing authority verification, and the prominent gold button for instant Gazette PIAR generation.

7. GENERAL JURY FAQ & PITCH ROOM BATTLECARD

WINNING EVERY QUESTION FROM ANY JUDGE

In open hackathon judging, jury members come from various domains. Here is how to answer the most common questions across different judge backgrounds:

TECH JUDGE "Doesn't the government e-procurement portal (NICGEP) already have built-in rules?"

Your Winning Answer:

"NICGEP is purely an *administrative filing cabinet*. It acts like an upload form—it validates that you uploaded a PDF, but it has **zero intelligence** to check whether the bidding window was compressed, whether a midnight corrigendum was unfair, or whether a single bidder is colluding. NICGEP processes whatever is uploaded; CHEIRAP acts as the intelligent sentinel that audits what is happening on NICGEP in real time."

FINANCE JUDGE "What is the actual Return on Investment (ROI) of this tool for the Government?"

Your Winning Answer:

"In India, once a rigged public contract is signed, the contractor immediately claims a **10% to 15% cash mobilization advance** from the state treasury. In our live monitoring of 92 Manipur tenders, CHEIRAP flagged **₹43 Crores of high-risk capex** before contracts were awarded. Intercepting those tenders prevents an immediate ₹4.3 to ₹6.4 Crores in untraceable cash leakage, while avoiding years of expensive High Court litigation."

UX / GOVTECH
JUDGE

"Government officers are notoriously resistant to complex new software. How do you get them to adopt this?"

Your Winning Answer:

"Line department engineers don't have to change anything! They continue using their standard NIC portal. CHEIRAP operates completely out-of-band for the **Vigilance Commission and Finance Department**. And for the vigilance officer, CHEIRAP saves hundreds of hours: instead of reading a 120-page tender document, CHEIRAP generates an instant 14-Section Gazette brief with the exact legal violation and ready-to-sign stay order text."

SECURITY JUDGE

"What if an emergency occurs (like a flood or landslide) and the government legitimately needs to hire a contractor in 48 hours?"

Your Winning Answer:

"We specifically accounted for that! Under General Financial Rules (GFR Rule 194) and Manipur PWD rules, emergency procurements have legal exemptions. CHEIRAP monitors whether an official *Disaster Management Certificate* or *Emergency Declaration* is attached. If verified, CHEIRAP applies an 'Emergency Exemption Verified' stamp and automatically downgrades the risk tier so legitimate emergency work is never stalled."

REGIONAL / POLICY

"Why is the project named CHEIRAP and why is the local script so important?"

Your Winning Answer:

"The **Cheirap Court** (ꯏꯝꯂꯩꯇ) was established in the 1570s at the historic Kangla Fort in Manipur as the highest royal court of justice and integrity. By naming our system CHEIRAP and implementing authentic **Meetei Mayek script** (ꯏꯝꯂꯩꯇ 𑜋𑜤𑜂𑜫 𑜇𑜨𑜃𑜫 𑜇𑜨𑜃𑜫 𑜇𑜨𑜃𑜫), we honor this heritage and demonstrate true localized GovTech that state officials can take cultural pride in deploying."

When presenting to a general jury panel, your pitch must be **fast, relatable, high-energy, and visual**. Follow this exact 180-second script while demonstrating the live application on screen:

TIME	WHAT YOU DO ON THE SCREEN	WHAT YOU SAY TO THE JUDGES
0:00 - 0:40	- Display Hero Screen with Manipur emblem and live surveillance ticker. - Point at the Capex meter showing ₹1,874 Crores monitored.	"Judges, imagine a bank that only checks its security cameras <i>two years after</i> a robbery occurred, when the cash is already gone. That is how government procurement auditing works today across India. By the time auditors write a report, crooked contractors have already received crores in upfront cash advances and abandoned the work. Meet CHEIRAP AI—named after Manipur's historic court of integrity. We stop procurement fraud in 48 hours, while bids are still sealed."
0:40 - 1:15	- Scroll smoothly into the Monitoring Dashboard. - Click the RED Tier filter showing 2 critical tenders.	"Right now, CHEIRAP is monitoring 92 real public tenders worth ₹1,874 Crores across 6 Manipur departments. Our Dual-Brain AI instantly filtered them into Green, Amber, and Red. Look at this Education Department tender for ₹4.82 Crores. An insider published the tender for just 3 days instead of 14, and snuck in a midnight technical change to disqualify honest competitors."
1:15 - 1:50	- Click the case to open the Case Detail Modal. - Click Tab 4: Explainable AI Waterfall.	"Why can't you just use ChatGPT for this? Because an AI chatbot hallucinates, and a judge cannot cancel a multi-crore contract based on a chatbot's feeling. CHEIRAP uses an Explainable AI Waterfall. It gives the officer an itemized receipt showing exactly which rule was broken: +28 points for window compression, +18 for late changes, and +12 for high deposits. It is 100% transparent and court-ready."
1:50 - 2:30	- Click the gold 'Gazette PIAR' button. - Scroll down the 14-Section Gazette view showing the emblem and cryptographic seal.	"Now, how does the officer take action? In one click, CHEIRAP generates an official 14-Section Pre-Award Integrity Assessment Report, formatted in the authentic style of The Manipur Gazette with Meetei Mayek script. It cites state Finance Department orders first, central rules second, and carries a tamper-evident SHA-256 digital seal."
2:30 - 3:00	- Click 'Print Official Gazette' to show the print preview dialog. - Return to hero screen for final verdict.	"With one tap, the Chief Vigilance Officer prints an official legal stay order to halt the tender before a single rupee leaves the treasury."

